

Is Gen AI Bad For The Environment?

Posted by u/JCunliffeUK • 0 points • 1 comments

Been having fun generating comics like the rest of the community but saw a couple of comments about how AI is damaging the environment.

I asked ChatGPT to do some deep research so I could be better informed and got the information below:

Comparison Table of Energy Usage and CO₂

Below is a summary table comparing the **electricity consumption** of generating one AI image to the various tasks discussed, along with approximate **carbon emissions** for each (assuming ~0.4 kg CO₂/kWh for illustration):

Activity	Energy Use (kWh)	Approx. CO ₂ Emitted
AI image generation (1 image)	~0.003 kWh [theverge.com](https://www.theverge.com/24066646/ai-electricity-energy-watts-generative-consumption#:~:text=watching%20nine%20seconds%20or%203,energy%20as%20charging%20your%20smartphone))	~1–2 grams CO ₂ (very low)
Charging a smartphone (1 full charge)	~0.012 kWh [theverge.com](https://www.theverge.com/24066646/ai-electricity-energy-watts-generative-consumption#:~:text=watching%20nine%20seconds%20or%203,energy%20as%20charging%20your%20smartphone))	~5–10 g CO ₂
Boiling a kettle (1 L water)	~0.1–0.15 kWh [theguardian.com](https://www.theguardian.com/notesandqueries/query/0,5753,-2452,00.html#:~:text=costing%20refers%20to%20NSW%20values,842cents%75%20g%20CO2))	~75 g CO ₂
Microwave use (5 minutes on high)	~0.12 kWh [siliconvalleypower.com](https://www.siliconvalleypower.com/residents/save-energy/appliance-energy-use-chart#:~:text=Range%2C%20electric%20Oven%202,06%20per%20hour))	~36–60 g CO ₂
Streaming video (1 hour, HD)	~0.8 kWh [iea.org](https://www.iea.org/commentaries/the-carbon-footprint-of-streaming-video-fact-checking-the-headlines#:~:text=Looking%20at%20electricity%20consumption%20alone%2C,8%20kWh))	~240–400 g CO ₂
Video gaming (console, 1 hour)	~0.2 kWh [gamesradar.com](https://www.gamesradar.com/how-much-does-it-cost-to-run-a-ps5#:~:text=,when%20playing%20games%20was%20203W))	~60–100 g CO ₂
Video gaming (gaming PC, 1 hour)	~0.3–0.5 kWh [energysage.com](https://www.energysage.com/electricity/house-watts/how-many-watts-does-a-computer-use#:~:text=Key%20Takeaways))	~90–250 g CO ₂
3D rendering (1 hour, mid-range PC)	~0.6–0.8 kWh [blog.render.st](https://blog.render.st/how-much-does-rendering-at-home-cost#:~:text=Rendering%20needs%20power%20,range%2C%20only%20expressed%20in%20Euro))	~180–400 g CO ₂
Dishwasher (per load, normal cycle)	~1.5 kWh (1–2 kWh range) [siliconvalleypower.com](https://www.siliconvalleypower.com/residents/save-energy/appliance-energy-use-chart#:~:text=Coffee%20maker%20brew%2C%20warmer%20on%200,05%20per%20use))	~450–750 g CO ₂
Washing machine (per load, warm wash)	~0.5–1.0 kWh (front-loader) [ubuildit.com](https://www.ubuildit.com/washing-machines-whats-in-the-spin-2#:~:text=Electricity))	~150–300 g CO ₂
Washing machine (per load, hot wash)	~2.0+ kWh (inefficient top-load) [ubuildit.com](https://www.ubuildit.com/washing-machines-whats-in-the-spin-2#:~:text=Electricity))	~600+ g CO ₂

(Table notes: CO₂ figures are rough estimates for illustration. Actual emissions depend on the electricity generation mix (which can range from near-zero for renewables to >800 g/kWh for coal). The “AI image” entry refers to a single inference on average; running many inferences or a longer AI session would scale up accordingly.)

I hadn't really seen any data up until this point. It seems to me like the generations themselves aren't too ecologically damaging.

Comments

u/AutoModerator • 1 points • 2025-04-16 10:07 UTC

Hey /u/JCunliffeUK!

If your post is a screenshot of a ChatGPT conversation, please reply to this message with the [conversation link] (<https://help.openai.com/en/articles/7925741-chatgpt-shared-links-faq>) or prompt.

If your post is a DALL-E 3 image post, please reply with the prompt used to make this image.

Consider joining our [public discord server](<https://discord.gg/hT6PXe8gdZ>)! We have free bots with GPT-4 (with vision), image generators, and more!

🤖

Note: For any ChatGPT-related concerns, email support@openai.com

I am a bot, and this action was performed automatically. Please [contact the moderators of this subreddit](/message/compose/?to=/r/ChatGPT) if you have any questions or concerns.